

Theorem 3: Polarization and Hodge Lattice Indices

the principal polarization and computability of block indices

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The «Dynamic Principle» Program — the hodge-laboratory

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1 Setting

The intersection form J on $H_1(C_N, \mathbb{Z})$ is not merely a topological invariant: it is a polarization of the weight-1 Hodge structure and defines the Hodge lattice indices per conductor blocks — integer invariants computable algorithmically.

Definition 1 (polarization). For closed forms $\tilde{E}([\alpha], [\beta]) = \iint_{C_N} \alpha \wedge \beta$; the polarization form on $H_1(C_N, \mathbb{Z})$ is the intersection form $E := J$.

Theorem 2 (J is a weight-1 polarization). *With respect to the Hodge decomposition $H^1 = H^{1,0} \oplus H^{0,1}$ the Riemann–Hodge relations hold: (1) $\tilde{E}(\omega, \omega') = 0$ for holomorphic forms; (2) $-\frac{1}{2i} \tilde{E}(\omega, \bar{\omega}) = \int_{C_N} |g|^2 dx dy > 0$. Hence $\text{Jac}(C_N)$ with J is a principally polarized abelian variety.*

Proof. (i) A curve has no nonzero holomorphic 2-forms, so $\omega \wedge \omega' = 0$ — the first Riemann relation read on the bilinear identity. (ii) $\omega \wedge \bar{\omega} = -2i|g|^2 dx \wedge dy$ gives $-\frac{1}{2i} \tilde{E}(\omega, \bar{\omega}) = \int_{C_N} |g|^2 dx dy > 0$. The unimodularity of J gives the principality. $\square$

Definition 3 (block indices). For a conductor $d \mid N$: the block lattice Λ_d — the isotypic component of characters with conductor d ; the saturation Λ_d^{sat} ; the block index $\text{Index}(\Lambda_d) = [\Lambda_d^{\text{sat}} : \Lambda_d]$. Block data: rank, signature of $E|_{\Lambda_d}$, determinant, index.

Theorem 4 (computability of the indices). *All quantities are finite and computed algorithmically from the closed forms: the saturation by Hermite–Smith reduction; the signature and determinant from the combinatorial intersection matrix filtered by characters; the index as a determinant ratio. The counts per block: $1 + 6 + 84 = 91$ ($N = 15$); $1 + 6 + 9 + 30 + 84 + 276 = 406$ ($N = 30$).*

Proof. Finiteness: $H_1(C_N, \mathbb{Z})$ is free abelian of rank $2g$. Computability: the isotypic component is defined over $\mathbb{Q}(\zeta_N)$; its intersection with the integer lattice is an integer sublattice; the saturation basis by the Hermite–Smith algorithm on the $\mu_N \times \mu_N$ -action table (sheet combinatorics). The restricted polarization: $E = J$, a combinatorial matrix. The census sums match the genera (V1). $\square$

Protocol data: numbers — certificate H

The stand polarization type (SNF): $(1, 1, 5, 5, 15, 15, 15, 15)$; the product of invariant factors = $1265625 = 3^4 \cdot 5^6 = 1125^2$ — exact integer agreement with the discriminant; $\text{vol}_h = 1125$.

Summary of results: summary

The polarization is proved (Riemann–Hodge); the computability is proved algorithmically; certificate H implements the computation with exact integer certification — the indices are not only computable but computed.

Verification: reproduction

python3 laboratory.py → “Certificates → H1”; Lean: snf_product, stand_disc.

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